Front-yard holiday LED display

Design: frame, W-modules, power fan-out

henderson (with Michael Mulhearn)

2026-10-19, phase 1

Inventory IDs in **bold sans** (e.g. **LED-1**) refer to the bill of materials in Section 7 and appear in the figures. The controller (Trenz Zynq module and level shifter) is described separately; this document stops at the data input of the first module.

1 Overview

- **Panel:** 56 × 37 square pixels at 33.3 mm (1.31 in) pitch, the pitch of a 30-LED/m strip, both across and down: an LED area of 73.5 in × 48.6 in, 2072 pixels.
- **Strip:** WS2815, 12 V, 30 LEDs/m, IP65 (**LED-1**); every LED individually addressable, with a backup data line (a dead LED does not darken the rest of the chain).
- **Modules:** 14 identical four-column *W-modules*, one 5 m reel each (148 of its 150 LEDs), plus one spare reel.
- **Power:** one sealed Mean Well HLG-600H-12B (**PS-1**, 12 V 40 A 480 W, IP67); one 5 A fuse and one 18 AWG home run per module.
- **Brightness cap:** 50 % in software (WLED or our own); up to about 75 % stays comfortable for the supply (Section 5).
- **Phase 1** is the full panel. A pilot (one module, Section 6) comes first.

2 Frame

The frame is a box of 2x6 on edge (**FR-1**), sized to the LED area plus clearance, with a 2x4 rail standing up across the inside at top and bottom (**FR-2**) and one 1x2 slat (**FR-3**) per two columns, screwed to the front of both rails (Figures 1 and 2).

- **Outside** about 77 in × 59 in × 5.5 in; inside 74 in × 56 in. Lumber: 8 ft boards only, scrap used elsewhere.
- **Rails** (**FR-2**): 2x4 standing up (3.5 in high, 1.5 in deep), set back from the front so that slat, strip and an optional diffuser fit within the box depth. The top rail area carries the connections; the bottom rail only holds the slats.
- **Slats** (**FR-3**): 28 1x2 (0.75 in × 1.5 in), each about 56 in long, at a pitch of two columns (66.7 mm); each carries two columns 33.3 mm apart, centered. The 10 mm strips overhang the 38 mm slat by about 3 mm each side, which is fine. A spacing jig on the rails keeps the pitch; gaps of about 29 mm leave the panel about 40 % open to wind and air.
- **Clips** (**MNT-1**): three per column (top, middle, bottom), screwed to the slat; 12 mm clips for the 10 mm strip in its IP65 coat.
- **Weather:** a rain cover with a drip edge over the top; all connections in the top rail area, under it. The supply (**PS-1**) is sealed and mounts on a backing plate behind the top rail; the fuse blocks and bus bars sit in a covered box beside it (**HD-5**).

- **Mounting:** on the low masonry wall in front of the house: a shoe over the wall cap, ropes from the top corners to the eaves (against falling forward), struts behind (against falling back); lay it flat in storms.
- **Measure first:** the frame's dimensions assume a 33.3 mm pitch. Measure the pilot reel over 37 LEDs before cutting the 2x6.

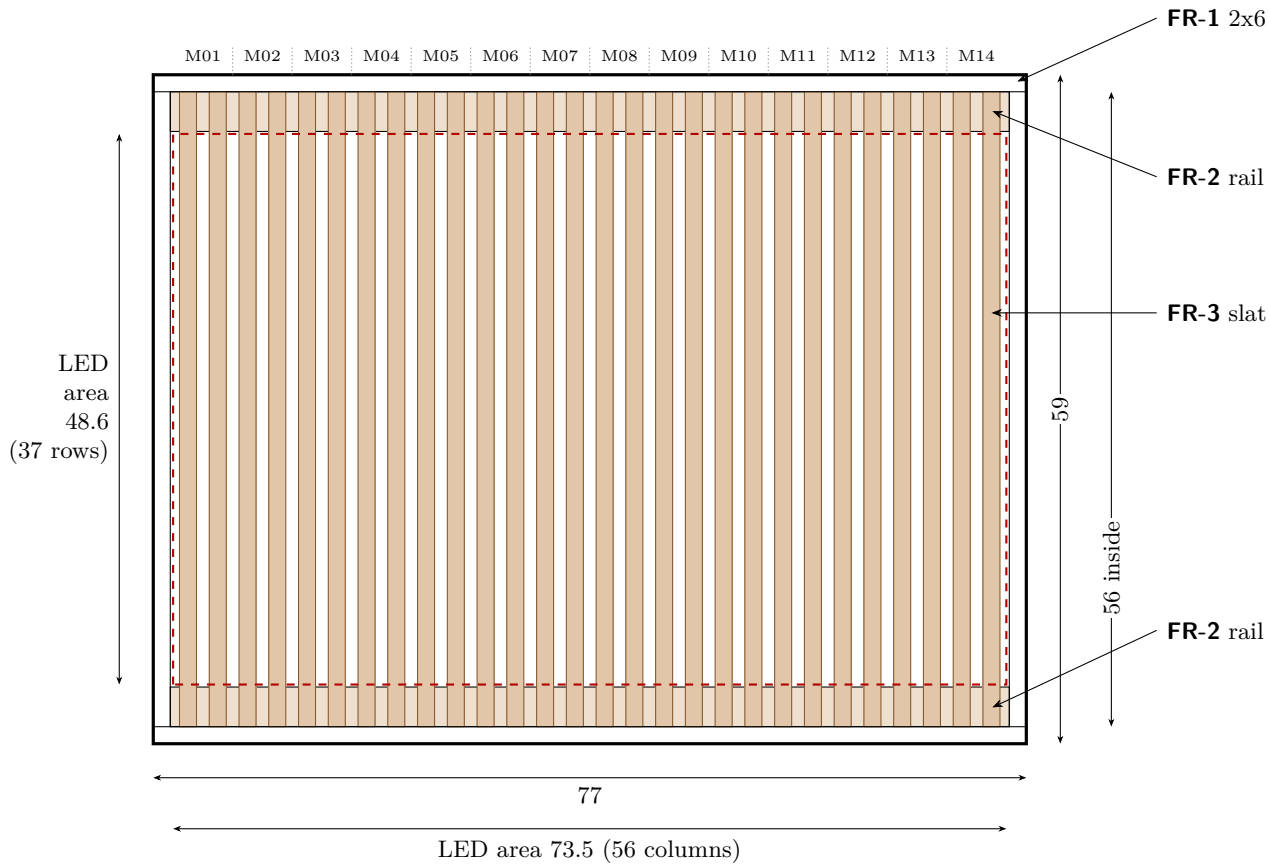


Figure 1: Frame, front elevation (to scale). 2x6 box **FR-1**, rails **FR-2**, 28 slats **FR-3** (two per module, M01–M14), LED area dashed. Dimensions in inches.

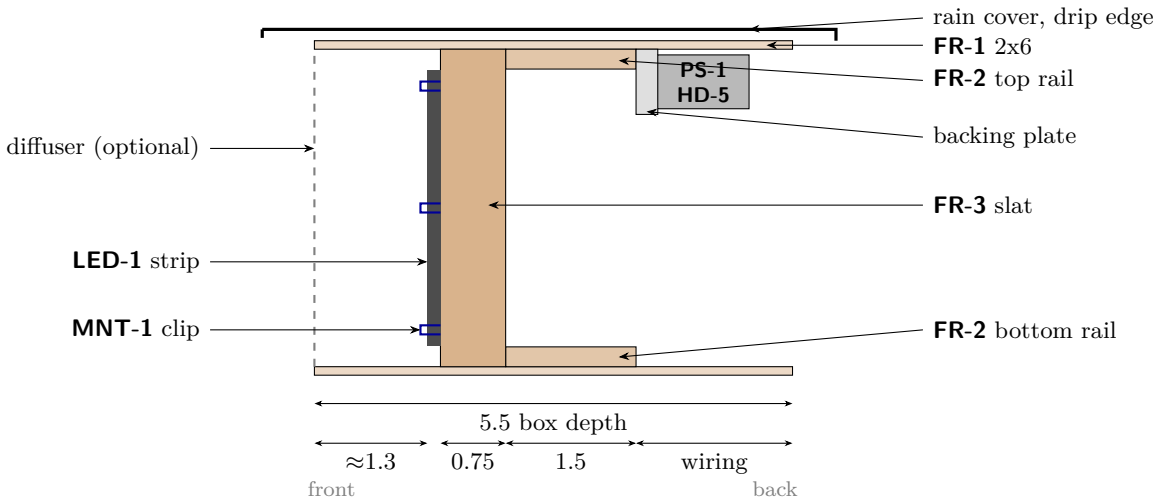


Figure 2: Frame, vertical section through a slat (not to scale; depths in inches, proposed). From the front: optional diffuser, air gap, strip **LED-1** in its clip **MNT-1**, slat **FR-3**, rail **FR-2**, then space for wiring behind. The supply **PS-1** and the covered box **HD-5** sit behind the top rail.

3 W-module

Each module is one reel (**LED-1**) folded into four columns of 37 LEDs (Figure 3):

- **Path:** the data enters at the top of column 1 and runs down; corner C1 (bottom) turns it up column 2; corner C2 (top) down column 3; corner C3 (bottom) up column 4; it leaves at the top of column 4. Input and output are both at the top, 100 mm apart.
- **Spare LEDs:** $4 \times 37 = 148$ of 150. The two spare LEDs are cut out at corner C1, so both factory ends, with their 4-pin plugs, stay on the module.
- **Corners:** each corner is cut at the strip's cut marks and joined by four short 18 AWG jumpers (**WIR-1**): 12 V, data (DO to DI), backup data (BO to BI) and ground (Figure 4). Each cut end is sealed with 1.25 in of adhesive-lined 1/2" heat shrink (**HS-1**): six per module.
- **Electrically** the module behaves as one uncut reel; the jumpers carry current better than the strip's own copper.
- **Power:** fed at the input end only, through the input plug's power leads (12 V constant-current LEDs tolerate the drop along the strip; checked on the pilot, and the far-end "voltage compensation" pigtail is in reserve).
- **Data:** modules plug together: module k's output plug into module $k{+}1$'s input plug. The first module's backup input BI is tied to ground.
- **Mounting:** the two slats of a module are its carrier; a module comes off with its clips.

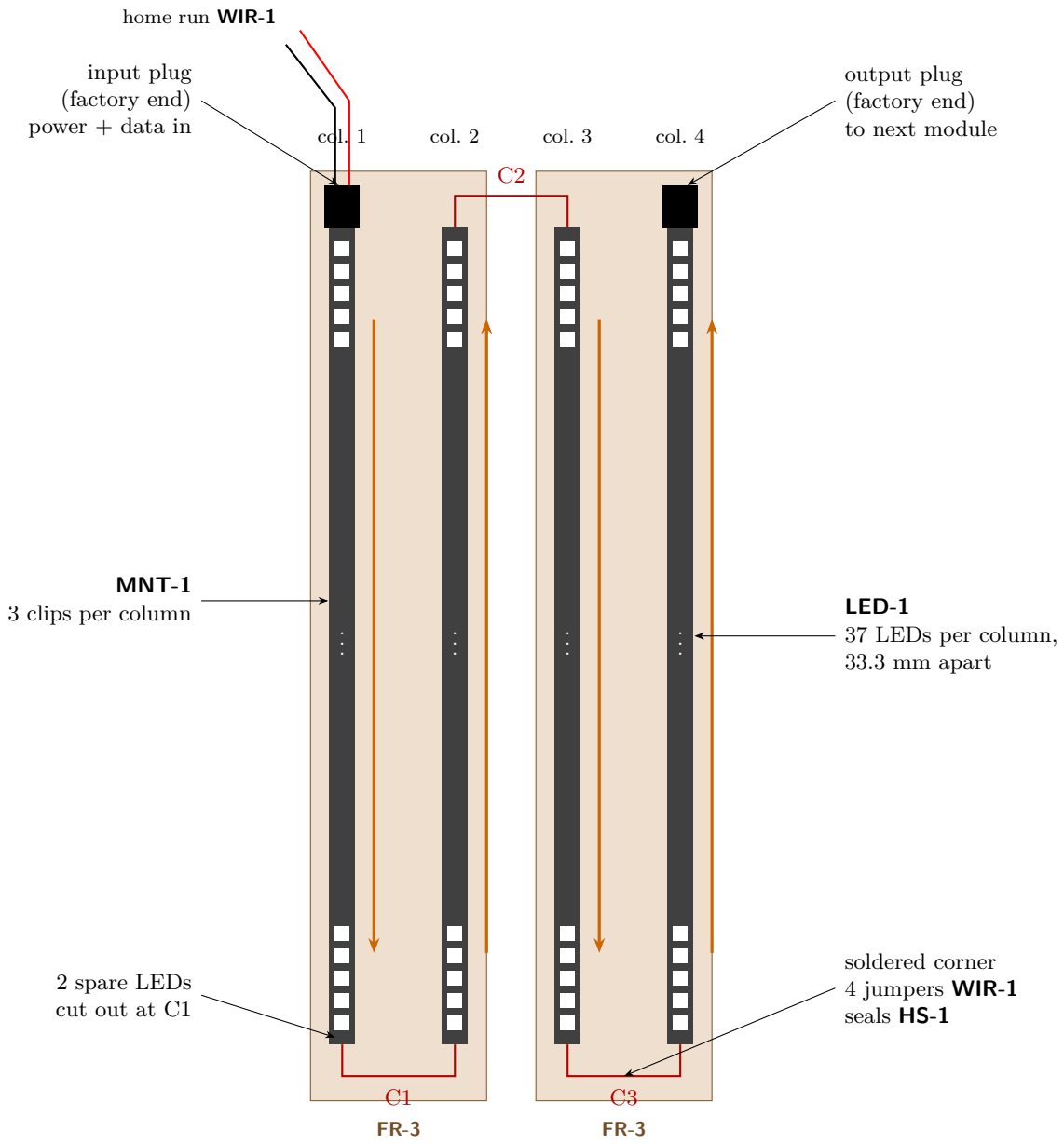


Figure 3: W-module: one reel **LED-1** as four columns of 37 on two slats **FR-3** (LEDs not all drawn). Arrows: data direction. C1–C3: soldered corners (Figure 4); the two spare LEDs are cut out at C1. Input and output are the reel's factory plugs; power enters at the input.

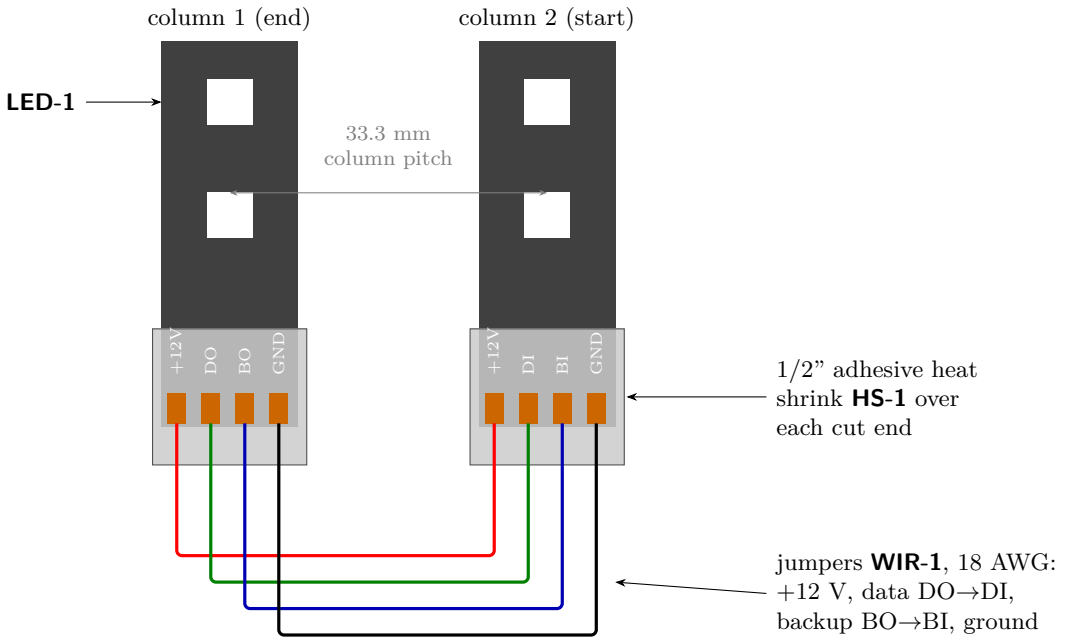


Figure 4: Corner joint (C1 shown). Four 18 AWG jumpers **WIR-1**: 12 V, data (DO to DI), backup data (BO to BI), ground. Each cut end sealed with 1/2" adhesive heat shrink **HS-1**.

4 Power fan-out

Figure 5 shows the power path; Figure 6 the layout along the top rail.

- **Mains:** a plug-ended cord (**HD-3**) to a GFCI outdoor outlet, spliced to the supply's AC leads in a weatherproof splice (**HD-4**). The house breaker protects the cord.
- **Supply PS-1**: HLG-600H-12B, 12 V 40 A 480 W, IP67, fanless, universal input 90 V to 305 V AC. Its dimming wires are unused: capped and insulated (open means full output).
- **Links** (**HD-1**): the supply's output to the two fuse blocks' + studs and the two bus bars' studs; heavy enough for the supply's current limit, so nothing upstream of the fuses needs its own fuse.
- **Fuse blocks FB-1** (two, 8 circuits each) with one 5 A ATO fuse **FU-1** per module: block A feeds M01–M07, block B M08–M14; one spare circuit each.
- **Ground bus bars BB-1** (two, 12 screws each): one per block.
- **Home runs:** one 18 AWG pair (**WIR-1**) per module, red from its fuse to the module's input plug power lead, black from there back to the bus bar; ring terminals **HD-2** at the blocks, soldered and heat-shrunk (**HS-2**) at the pigtail. Runs are at most about 1.5 m.
- **Protection by layers,** each guarding the wiring after it: the house breaker (cord); the supply's own current limit (links); a 5 A fuse (each home run and module).
- **Rule:** switch off before plugging or unplugging a module (there are no bulk capacitors).

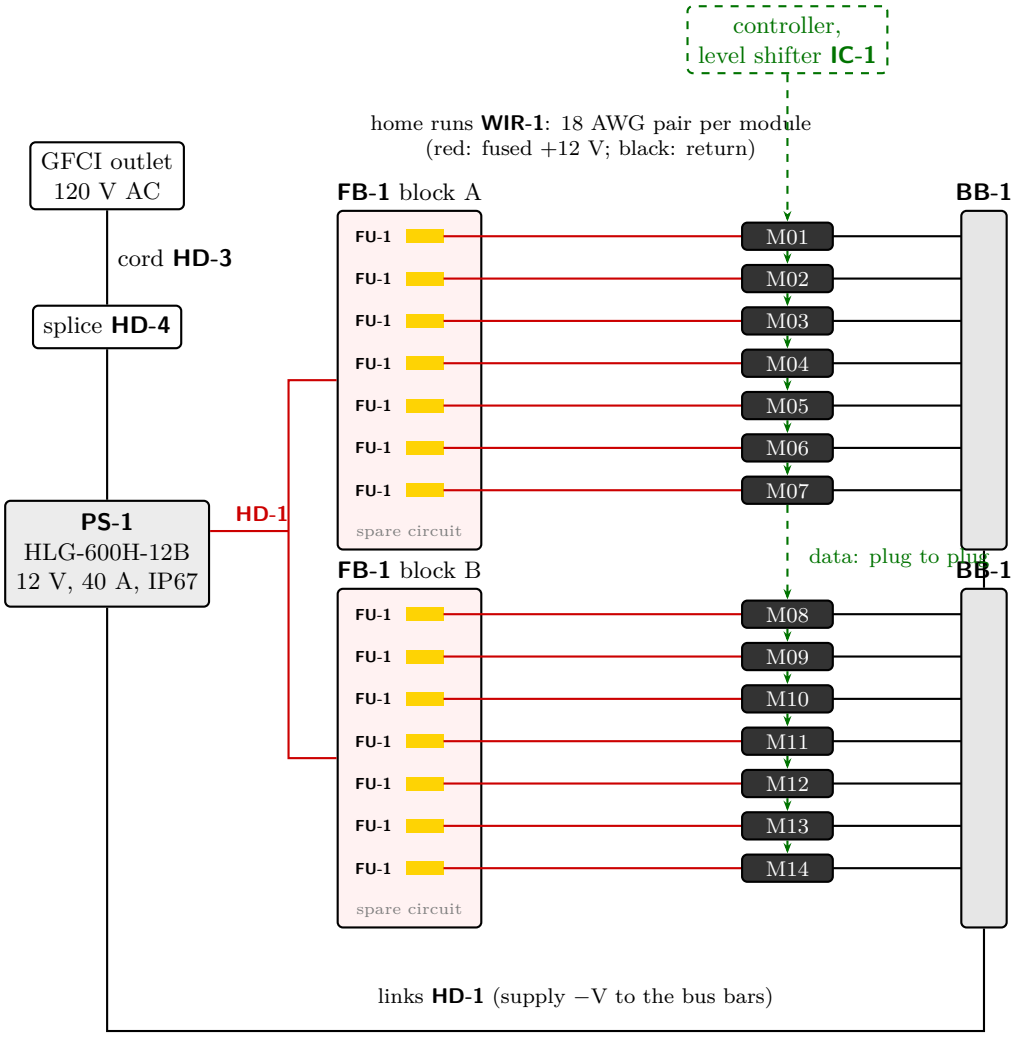


Figure 5: Power fan-out. Mains cord **HD-3** and splice **HD-4** to the supply **PS-1**; links **HD-1** to two fuse blocks **FB-1** with 5 A fuses **FU-1** and two ground bus bars **BB-1**; one 18 AWG home run **WIR-1** per module to its input plug. Data (dashed) runs from the controller's level shifter **IC-1** through the modules' plugs.

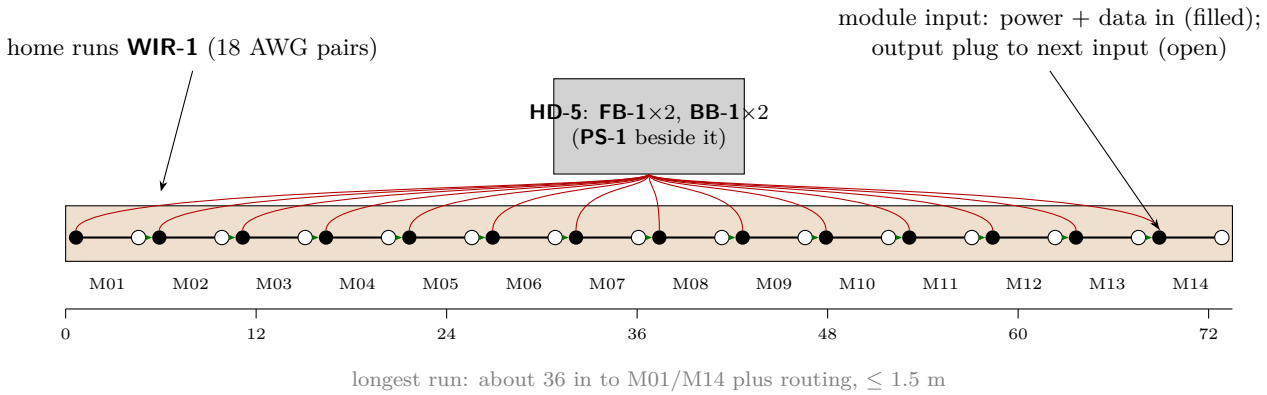


Figure 6: Top rail, seen from the front: module inputs (filled) and outputs (open), the plug-to-plug data chain, and the home runs **WIR-1** from the covered box **HD-5** (fuse blocks **FB-1**, bus bars **BB-1**) behind the middle of the top rail. Horizontal scale in inches from the LED area's left edge.

5 Numbers

BTF-LIGHTING rates a reel at 12 V 3 A (36 W) at full white; a module uses 148 of its 150 LEDs.

case	per module	panel at 12 V	panel power	supply load
full white, 100 %	3.0 A	42 A	504 W	105 % (over)
75 % cap	2.2 A	32 A	378 W	79 %
50 % cap (design)	1.5 A	21 A	252 W	52 %
typical scene, 50 %	< 0.5 A	< 6 A	< 70 W	< 15 %

- **At the wall:** about 280 W at the 50 % cap, about 2.5 A at 120 V.
- **Fuses:** 5 A is the next standard size above 125 % of 3 A; it never blows in normal use and protects 18 AWG.
- **Home-run drop:** 1.5 m of 18 AWG out and back is about 0.06 Ω: 0.1 V at 1.5 A, 0.2 V at 3 A.
- **Along a module** (fed at one end): about 1 V at the 50 % cap, about 2 V at full white (estimate); WS2815 LEDs are constant-current and hold their color to about 10 V. To be measured on the pilot.
- **Data:** one chain of 2072 pixels takes about 62 ms per frame (about 16 frames per second); splitting into more chains, or one line per module from the Zynq, is a matter of unplugging joints.

6 Pilot

One reel, ordered 2026-10-19 (Amazon order 1). Before cutting: light the whole reel from a lab 12 V supply with current limiting (start at 1 A; about 3.5 A for full white), check every LED and the color order, and measure the far end. Then: measure the pitch over 37 LEDs, fit a clip, cut and solder the three corners, seal, mount on two slats, and repeat the test, including the voltage at the far end at the 50 % cap.

7 Bill of materials

Prices as read 2026-10-18/19, before shipping and tax. Order: P = Amazon pilot (placed), A2 = Amazon order 2 (after the pilot), DK = DigiKey order v1, HD = Home Depot (when laying out), OH = on hand. Links: `https://www.amazon.com/dp/<ASIN>`.

ID	item	qty	order	source	part / ASIN	total
LED-1	BTF-LIGHTING WS2815, 12 V, 5 m, 150 LEDs, IP65, gold wire, backup data	1+14	P, A2	Amazon	B07LG54F4W	$449.85
MNT-1	BTF-LIGHTING transparent strip clips, 12 mm, 100 + screws per pack	2	P	Amazon	B0DM8VL3R4	$27.98
WIR-1	TUOFENG 18 AWG silicone wire, tinned copper, 50 ft red + 50 ft black per pack	1+1	P, A2	Amazon	B07FMWNF3J	$33.98
HS-1	XHF heat shrink 1/2”, 3:1 adhesive-lined, black, 10 ft	1	P	Amazon	B0919K23RX	$6.29
HS-2	XHF heat shrink 3/16”, 3:1 adhesive-lined, black, 20 ft	1	P	Amazon	B0919F4YFZ	$7.99
IC-1	SN74AHCT125N, DIP-14, generic 10-pack (pilot, spares)	1	P	Amazon	B08R6BCSYC	$7.99
IC-2	SN74AHCT125N, Texas Instruments (installation)	5	DK	DigiKey	296-4655-5-ND	$5.00
PS-1	Mean Well HLG-600H-12B, 12 V 40 A 480 W, IP67	1	DK	DigiKey	1866-2567-ND	$209.00
FB-1	Fuse block, 8 circuits, ATO/ATC, #8-32 outputs, #10-32 stud, cover	2	A2	Amazon	B0F7P88MZ2	$47.98
BB-1	RVBOATPAT ground bus bar, 12×#8 screws, 1/4” stud, cover	2	A2	Amazon	B0F6SC5C3Q	$19.96
FU-1	Littelfuse 0287005.PXCN, 5 A ATO fuse	20	DK	DigiKey	F4197-ND	$6.66
FR-1	2x6, 8 ft (two cut 77”, two 56”)	4	OH	lumber		–
FR-2	2x4, 8 ft (two rails, about 74”)	2	OH	lumber		–
FR-3	1x2, 8 ft (slats, 56” each; 28 + 2 spare)	30	OH	lumber		–
HD-1	Links, supply to blocks and bus bars, 12–10 AWG, ring terminals (#10, 1/4”)	–	HD	Home Depot		–
HD-2	Ring terminals, #8, 22–16 AWG, heat-shrink sealed (about 50)	–	HD	Home Depot		–
HD-3	Plug-ended outdoor cord, 3 conductors (or one on hand)	1	HD	Home Depot		–
HD-4	Weatherproof splice for the supply’s AC leads	1	HD	Home Depot		–
HD-5	Covered box for fuse blocks and bus bars	1	HD	Home Depot		–

Totals: Amazon pilot $97.23 (placed); Amazon order 2 $504.79; DigiKey v1 $220.66; together $822.68, plus Home Depot items.